2- AND 3-YEAR-OLDS COORDINATE DECISIONS IN DYADS & TRIADS

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BACKGROUND
Ontogeny of Cooperation

16 months Dual cooperation
(with one other peer)



18 months Triadic conversations
(with a parent and sibling)



AGE 3

There are no studies on **children younger than 5's ability to coordinate decisions in groups**, although we suspected they might develop the capacity **around age 3.**

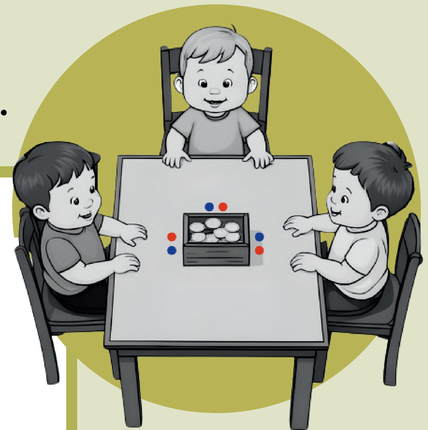
5-Years-Old Group cooperation
(with coordination and an understanding of roles)



Our ability to **coordinate decisions** and **generalize** our coordination to other **domains** distinguishes our cooperation from that of other species.

MAIN TAKEAWAY

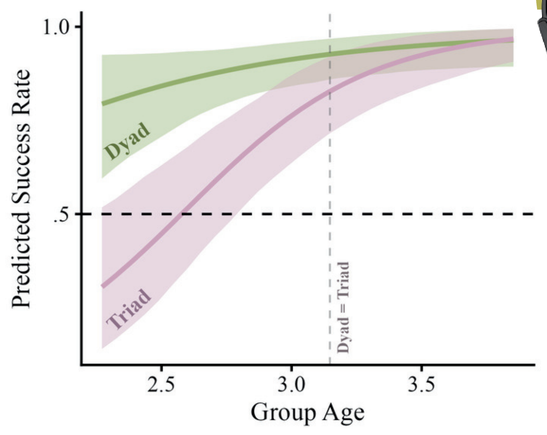
By age 3, children can coordinate decisions in groups.



When can children cooperate in peer groups?
Age 3!*

100% probability of a strong interaction between age and group size on success

Children's success across the 3 test trials.



*A Johnson-Neyman test indicated that children succeed at similar rates in dyads and triads by age 3.15!

Task

Children learned that the box would unlock when each partner used the same color key (either red or blue). Children could not see each other's chosen color.

METHODS

2x2 between-subjects design

Age \ Group Size	2-Year-Old Dyad	2-Year-Old Triad
	3-Year-Old Dyad	3-Year-Old Triad

dyad = pair 

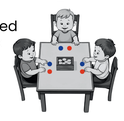
triad = group of 3 

PROCEDURE

Individual Training Group Warm-Up

1 Individual Training
Experimenter(s) taught each child individually how the box works.

2 Group Warm-Up
Children practiced opening the box together, once.



3 3 Test Trials
Children opened the box three more times (with increasingly informative prompts).



4 Domain Generalization Trial
In the final trial, children were given pig and cow, instead of red or blue, with no additional instruction.

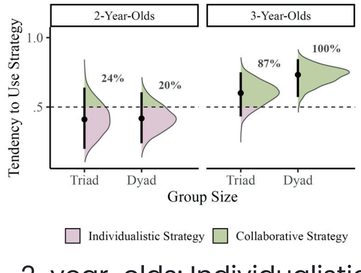


RESEARCH QUESTIONS:

- **When can children coordinate decisions** in peer groups?
- What approach do children take to coordinate?
- Can children generalize their coordination?

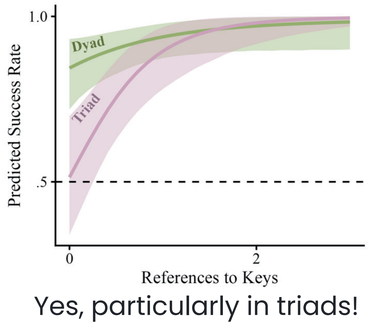
ADDITIONAL FINDINGS

What approach did children take to opening the box?



2-year-olds: Individualistic
3-year-olds: Collaborative

Did communication aid coordination?



Yes, particularly in triads!

When can children generalize their coordination to a new domain?
(by matching on pig or cow instead of red or blue)



	Dyads	Triads
2-year-olds	10	1
3-year-olds	10	10

By age 2 in dyads and age 3 in triads!

PARTICIPANTS

N = 100
10 groups per condition

50% female
26% URM

Children were tested either in the lab or their daycare in Durham, NC.

DISCUSSION

By 3 years of age, children can coordinate decisions with more than one other peer. Because 3-year-olds rely on collaborative strategies such as communication to coordinate, they can also generalize this coordination to new domains in groups.

These findings support the **interdependence hypothesis**, which posits that cooperation evolved in two stages.

phylogeny, 

ontogeny 

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